

# Require a non-throwing default contract-violation handler

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## Abstract

This paper proposes a one-line addition to require that the system-supplied default violation handler not throw.

## 1 Introduction

Some people are concerned that a new codepath might result if an *implicit* precondition — a precondition attached by the implementation to a core language expression, as proposed in [P3081R1], [P3100R1], and [P3329R0] — is violated and the contract-violation handler exits via an exception.

This paper addresses that problem by requiring that the *default* contract-violation handler does not exit via an exception. Thus, the new code path will not happen without an explicit replacement of the contract-violation handler by the user, which means it cannot happen inadvertently by accident or as the result of a port to a conforming implementation.

Note that this restriction can always be relaxed in a later iteration of C++ as a non-breaking change, but it cannot be so added.

## 2 Proposed wording

The proposed wording is relative to [P2900R13].

Modify [basic.contract.handler], paragraph 1 as follows:

The *contract-violation handler* of a program is a function named `::handle_contract_violation`. The contract-violation handler shall take a single argument of type “lvalue reference to `const std::contracts::contract_violation`” and shall return `void`. The contract-violation handler may have a non-throwing exception specification. The implementation shall provide a definition of the contract-violation handler, called the *default contract-violation handler*. The default contract-violation handler shall not exit via an exception. [ *Note*: No declaration for the default contract-violation handler is provided by any standard library header. — *end note* ]

### 3 Summary

By adding this one line of wording, we eliminate a stated concern, and leave open all options moving forward. It is anticipated that this proposal will increase consensus, as it has no obvious or likely negative consequences.

### Acknowledgements

Thanks to Timur Doumler for providing the wording.

### Bibliography

- [P2900R13] Joshua Berne, Timur Doumler, and Andrzej Krzemiński. Contracts for C++. <https://wg21.link/p2900r13>, 2025-01-13.
- [P3081R1] Herb Sutter. Core safety profiles for C++26. <https://wg21.link/p3081R1>, 2025-01-06.
- [P3100R1] Timur Doumler, Gašper Ažman, and Joshua Berne. Undefined and erroneous behaviour is a contract violation. <https://wg21.link/p3100r1>, 2024-10-16.
- [P3329R0] Timur Doumler and Joshua Berne. Making erroneous behaviour consistent with Contracts. <https://wg21.link/p3329r0>, 2025-01-13.